

AI Response Validation System

Full Project Title: Development of AI Response Validation System with Hallucination Detection Assistance (Group 1)

Short Name: AI Response Validation System

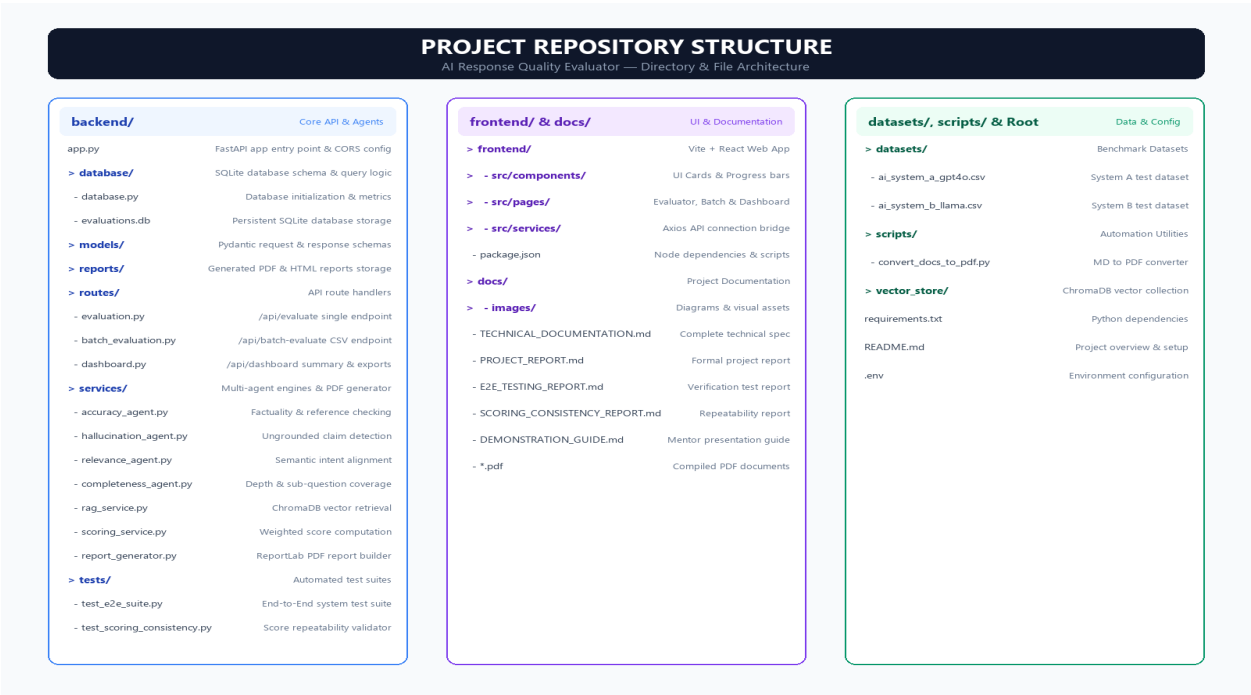
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A production-grade, multi-agent evaluation platform for assessing generative AI model responses using RAG evidence retrieval, automated metric scoring, real-time analytics dashboard, and executive PDF report exports.

Key Features

- * **Multi-Agent Evaluation Framework:** Evaluates responses across **Accuracy**, **Hallucination**, **Relevance**, and **Completeness**.
- * **RAG-Powered Evidence Grounding:** Vector retrieval via **ChromaDB** checks facts against authentic knowledge sources.
- * **Batch Evaluation Pipeline:** Upload CSV files containing bulk prompt-response datasets for automated evaluation.
- * **Interactive Analytics Dashboard:** Real-time pass/fail rates, dimension averages, quality trends, and multi-filter criteria.
- * **Executive PDF & HTML Export:** Programmatic report generation built with ReportLab featuring charts, flagged hallucination analysis, and recommendations.
- * **Multi-Model Benchmark Comparison:** Evaluate and compare System A vs System B performance side-by-side.

Project Structure



```
# Activate virtual environment venv\Scripts\activate # Install dependencies pip install -r requirements.txt # Start FastAPI server uvicorn backend.app:app --reload --port 8000
```

2. Start Frontend Web App

```
cd frontend npm install npm run dev
```

Open <http://localhost:5173> in your browser.

■ Running Unit & E2E Tests

```
# Run End-to-End Test Suite python -m unittest backend/tests/test_e2e_suite.py # Run Scoring Consistency Validation python -m unittest backend/tests/test_scoring_consistency.py
```

■ Documentation Links

* ■ [Technical Documentation](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/docs/TECHNICAL_DOCUMENTATION.md)

* ■ [Formal Project Report](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/docs/PROJECT_REPORT.md)

* ■ [End-to-End Testing Report](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/docs/E2E_TESTING_REPORT.md)

* ■ [Scoring Consistency Report](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/docs/SCORING_CONSISTENCY_REPORT.md)

* ■ [Mentor Demonstration Guide](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/docs/DEMONSTRATION_GUIDE.md)